



02451 Introduction to Machine Learning

Week 13: Recap and discussion of the exam

Bjørn Sand Jensen

5 May 2026

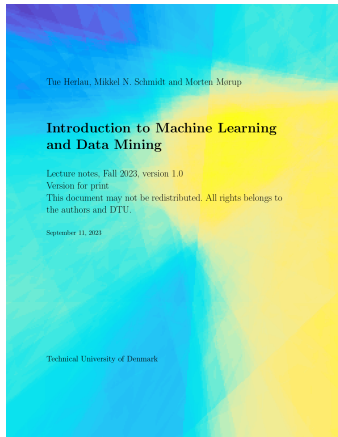
DTU Compute, Technical University of Denmark

Today

Feedback Groups of the day:

Anes Emad Al-Kaisi, Simon Einar Boas, Mikkel Cornelius Buch-Hansen, Albert Bernhard Bunkenborg, Chandler Desforges, Maryam Dowlatabadian, Lærke Djurhuus Ipsen, Albert Hemming Koba, Ninja Lykke Laursen, Rei Ern Joy Lee, Martim Pacheco Ferreira Prezado Louro, Louise Vomb Boje Lund, Angel Molina Guillen, Catarina Neto Anastácio, Jørn Kiil Nielsen, Laura Toft Nielsen, Berat Ege Percinel, Mathias Petersen, Philipp Ruf, Monica Santinelli, Annabelle Stahr, Felis Le Yao Tan, Keshav Taparia, Adrian Vexø van Arkens, Rosemarie van der Watt Wodschow, Kalle Yttesen

Reading/homework material: Chapter 1-20



Lecture Schedule

- 1 Introduction, data, and visualization
3 February: C1, C2, C7
- 2 Summary statistics, similarity, and nearest neighbors
10 February: C4, C12
- 3 Computational linear algebra and PCA
17 February: C3
- 4 Probability theory, Bayes and Naive Bayes
24 February: C5, C6, C13 (Assignment 1-3 due on 26 February)
- 5 Bayesian inference and linear models
3 March: C5, C6, C8
- 6 Decision trees, overfitting and cross-validation
10 March: C9, C10
- 7 Performance evaluation and AUC
17 March: C11, C16 (Assignment 4-6 due on 19 March)
- 8 Artificial Neural Networks and Optimization
24 March: C15
- 9 Bias/Variance and ensemble methods
7 April: C14, 17
- 10 K-means and hierarchical clustering
14 April: C18 (Assignment 7-9 due on 16 April)
- 11 Mixture models and density estimation
21 April: C19, C20
- 12 Representation learning
28 April: Material provided separately
- 13 **Recap and discussion of the exam**
5 May: C1-C20 (Assignment 10-12 due on 7 May)

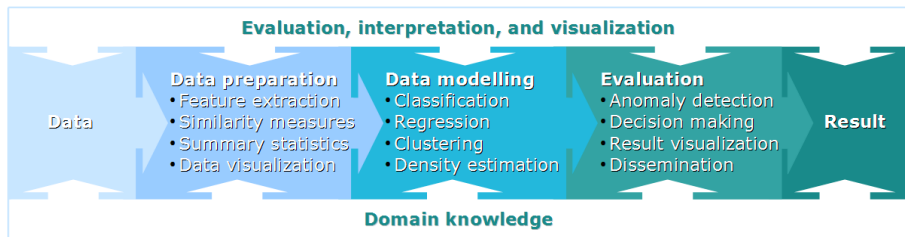
Online help: Piazza

Videos of lectures: <https://panopto.dtu.dk>

Streaming of lectures: Zoom (link on DTU Learn)

Recap of Week 12

Today's Learning Objectives



Learning Objectives

- Remember key aspects taught in the course
- Practical information about the exam

What is Machine Learning (ML)?

Alan Turing(1946)

We are not in a position to answer if a machine can think because the terms machine and think are undefined. Rather we should ask if a machine can imitate a human (the Turing test).

Proposed we should consider machines that were able to learn like children.



Alan Turing
(1912-1954)

Arthur Samuel (1959)

Machine learning: *Field of study that gives computers the ability to learn without being explicitly programmed*

Samuels wrote a checkers playing program, had the program play 10000 games against itself and work out which board positions were good and bad depending on wins/losses.



Arthur Samuel

Tom Mitchell (1999)

Well posed learning problem: *A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .*

For checkers we have:

E = 10000 games

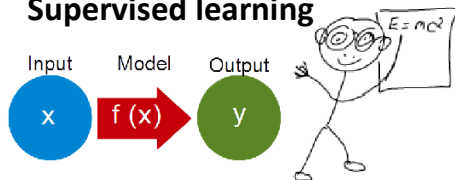
T = Playing checkers

P = If you win or not



What can ML be used for?

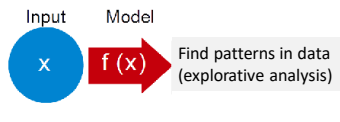
Supervised learning



y: different classes
y: continuous values

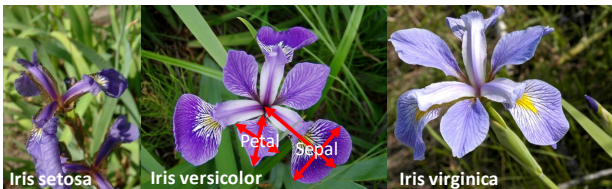
⇒ classification problem
⇒ regression problem

Unsupervised learning



Density estimation
Association mining
Clustering

Example: Fisher's Iris data



https://en.wikipedia.org/wiki/Iris_flower_data_set#/media/File:Kosaciec_szczecinkowaty_Iris_setosa.jpg

https://en.wikipedia.org/wiki/Iris_flower_data_set#/media/File:Iris_versicolor_3.jpg

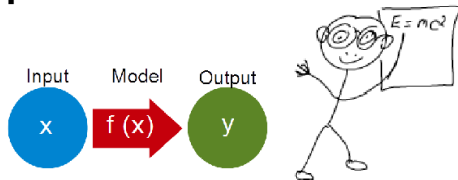
https://en.wikipedia.org/wiki/Iris_flower_data_set#/media/File:Iris_virginica.jpg



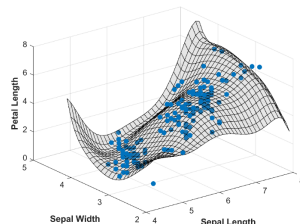
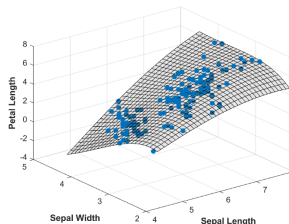
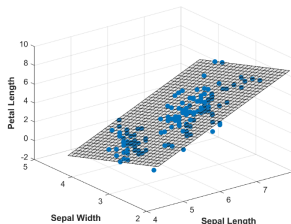
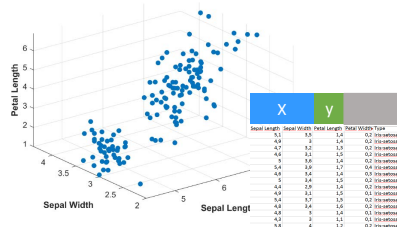
Ronald Fisher
(1890 - 1962)

Sepal Length	Sepal Width	Petal Length	Petal Width	Type
5,1	3,5	1,4	0,2	<u>Iris-setosa</u>
4,9	3	1,4	0,2	<u>Iris-setosa</u>
4,7	3,2	1,3	0,2	<u>Iris-setosa</u>
4,6	3,1	1,5	0,2	<u>Iris-setosa</u>
5	3,6	1,4	0,2	<u>Iris-setosa</u>
5,4	3,9	1,7	0,4	<u>Iris-setosa</u>
4,6	3,4	1,4	0,3	<u>Iris-setosa</u>
5	3,4	1,5	0,2	<u>Iris-setosa</u>
4,4	2,9	1,4	0,2	<u>Iris-setosa</u>
4,9	3,1	1,5	0,1	<u>Iris-setosa</u>
5,4	3,7	1,5	0,2	<u>Iris-setosa</u>
4,8	3,4	1,6	0,2	<u>Iris-setosa</u>
4,8	3	1,4	0,1	<u>Iris-setosa</u>
4,3	3	1,1	0,1	<u>Iris-setosa</u>
5,8	4	1,2	0,2	<u>Iris-setosa</u>

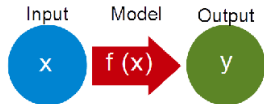
Supervised learning - regression



y: continuous values $\Rightarrow$ regression problem

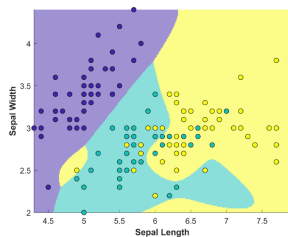
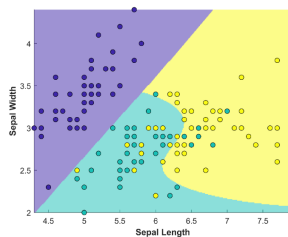
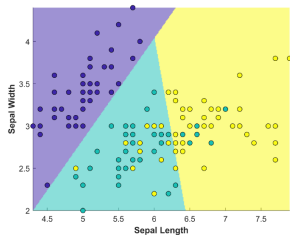
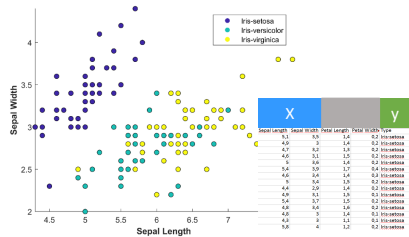


Supervised learning - classification

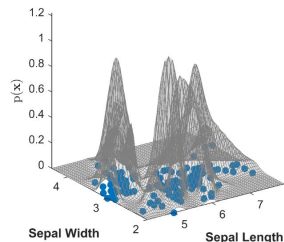
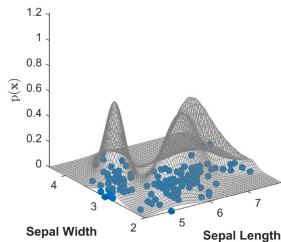
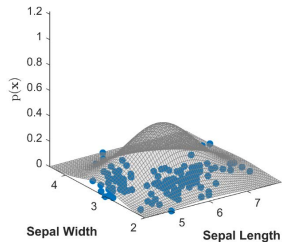
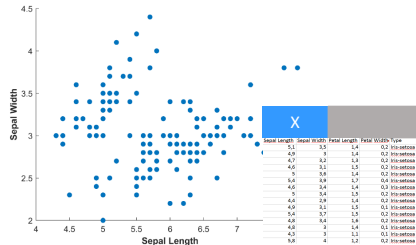
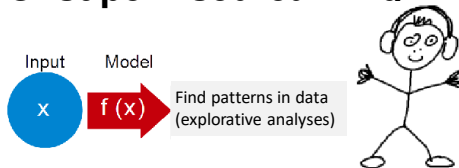


y: different classes

⇒ classification problem

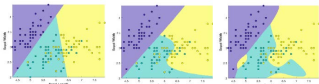


Unsupervised learning

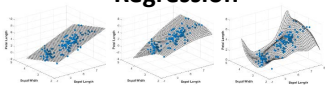


Control of model complexity

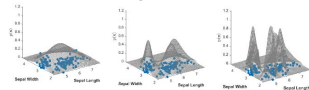
Classification



Regression



Density estimation



Adequate complexity



Too simple

Too complex



William of Ockham
(1288-1347)

*Lex Parsimoniae,
Law of parsimony*
Given two models with same
predictive performance, the
simpler model is preferred over
the more complex model
(paraphrased)



Albert Einstein
(1879 - 1955)

*"Everything should be made
as simple as possible, but not
simpler"*

Machine learning in one sentence

The aim of machine learning is to minimize the generalization error.

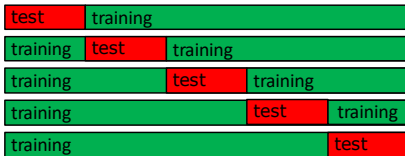
Generalisation error: The extent to which a machine learning method on average will fail when evaluated on an infinite amount of previously unseen data.

Cross-validation: A framework for quantifying the generalization error from the available data.

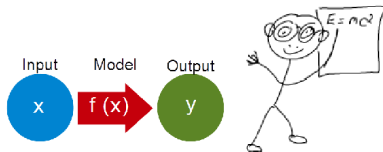
Hold – out method



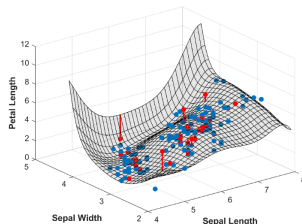
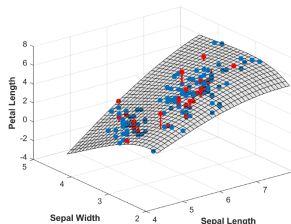
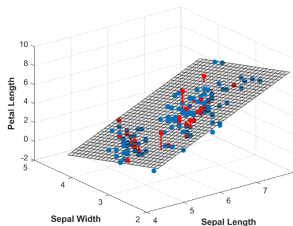
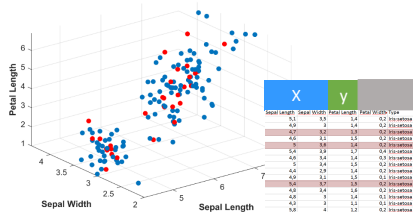
K-fold cross-validation (K=5)



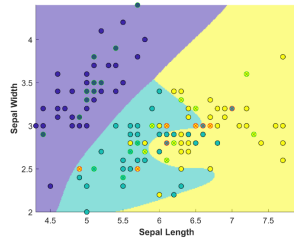
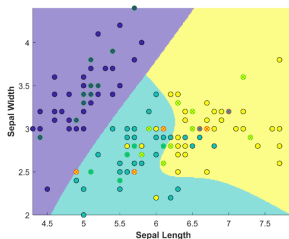
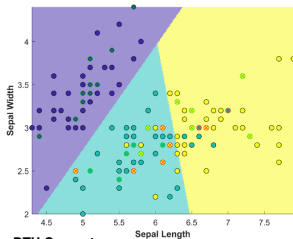
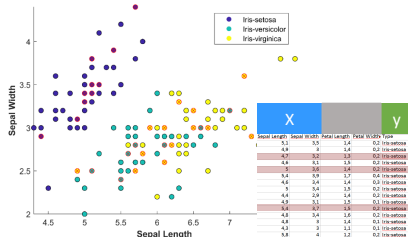
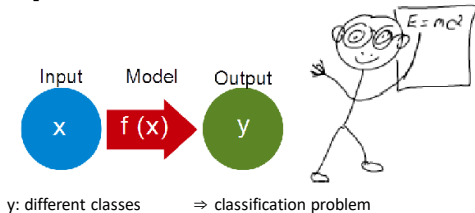
Supervised learning - regression



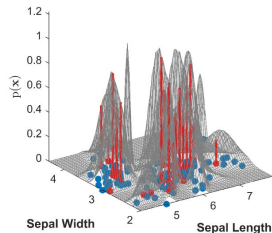
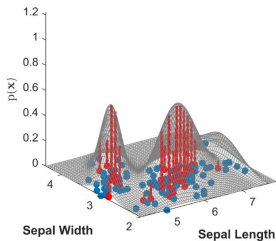
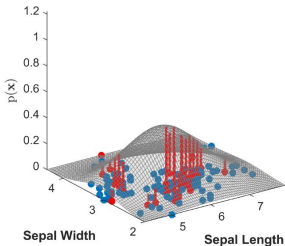
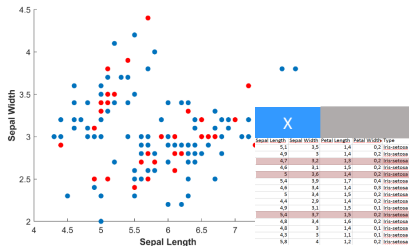
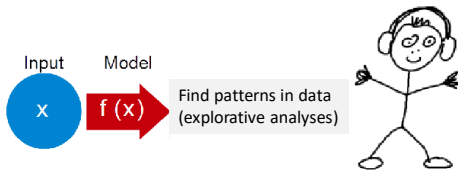
y : continuous values ⇒ regression problem



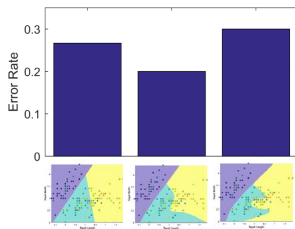
Supervised learning - classification



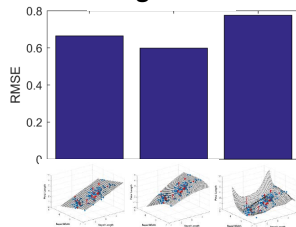
Unsupervised learning



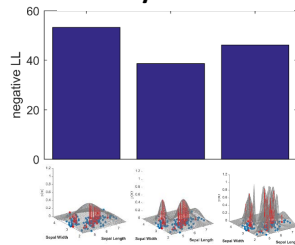
Classification



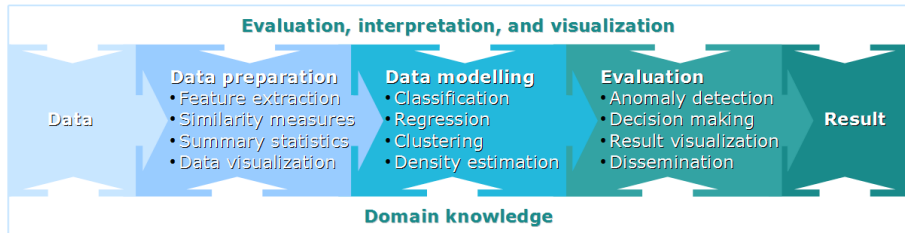
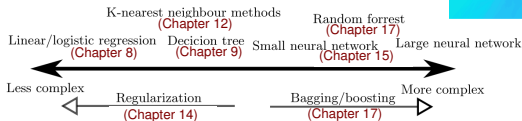
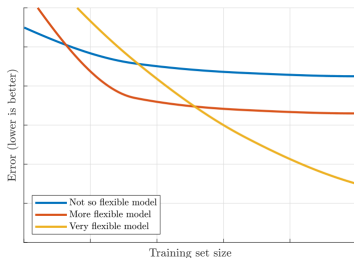
Regression



Density estimation



The Machine Learning workflow



Feedback/Improvements

Context: The most significant course update in eight years!

- Course overall: "I have learned a lot!" vs. "sometimes too abstract...", "well organised" vs. "assignment not relevant for exam..."
- Lectures/book: "Too much math" vs. "too little math". "Need more focus on practical aspects/implications"
- Assignments (brand new this semester): "very nice/useful" vs. "too much/difficult programming", "a few annoying bugs", "feedback could be made available sooner", "should be project work".
- Workload too high for a 5 ECTS course (9 hours per week on average)...
- Exam format is "outdated" (not aligned with the assignments)
 - 02270, 02323, 02402, 02403, 02451, 02452, 02456, 02516, 02517, ...
- Videos/recordings

Ideas welcome!

What should I do if I want more Machine Learning?

This course (02451) is a recommended prerequisite for:

Core ML courses:

- (02462 Signals and Data - KID students only)
- 02456 Deep learning
- 02465 Reinforcement learning
- 02471 Machine learning for signal processing
- 02476 Machine Learning Operations
- 02477 Bayesian machine learning
- 02460 Advanced Machine Learning (require: 02451/02452, 02456, 02476, 02477, 02405)

Special courses (typ. on advanced/research topics in core/applied ML).

Projects: BSc, MSc and PhD (see e.g. <https://cogsys.pages.compute.dtu.dk/projects/>)

Assessment

The final grade is based on an **overall assessment** of the assignments (approx. 1/10) and the written exam (approx. 9/10).

The assignments: 8 out of 12 assignments **must** be **approved/passed** to pass the course! All counts towards the final grade.

The written exam:

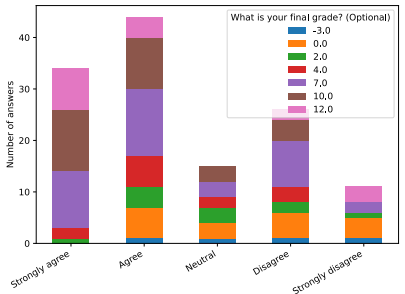
- **When:** 30th May 2026 (09:00 - 13:00) (see <https://eksamensplan.dtu.dk>).
- **Location:** DTU Campus (see <https://eksamensplan.dtu.dk>).
- **Duration:** 4 hours.
- **Exam set:** Available on site in physical paper form!
- **Format:** **No aids!**
 - **Not allowed:** course material, old exam sets, computer, calculators, tablet, phone, etc.
 - **Allowed:** two A4 sheets of handwritten notes (front and back). Printed notes are not allowed!
- **Type:** Multiple choice with 5 options and 1 correct answer. Correct answer: +3; incorrect: **-1**, don't know: 0).
- **Syllabus:** See "Course handbook" document or the public webpage.

Physical pen and paper exam

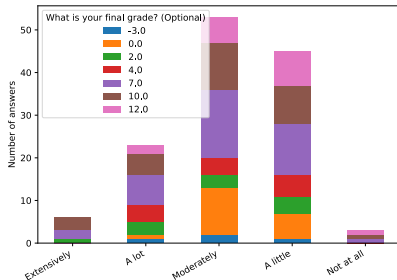
- Exam set available and printed on site.
- Hand-in the **provided** "Answer sheet"
- Write clearly: student ID "s123456" and fill in the associated bubbles below the individual numbers.
- Write clearly: Room name./no. and table no. See <https://eksamensplan.dtu.dk/>
- Select only one answer per question by clearly filling in the associated circle.
- Do **not** provide explanations, computations/rationale for the answers.
- Provide **only one** answer per question: A, B, C, D, otherwise mark E ("Don't know")!
- We will likely **permute** the options (i.e. A, B, C,...).
- Advice: Use a black pencil and a rubber (drawing compass).

[illegible]

Exam format: some data



Did the preparation of the handwritten notes help you better understand the materia



To what extent did you use your notes during the exam to solve the questions?

... a few more comments/suggestions

- Before starting: Skim/read the whole exam set and prioritize questions you are comfortable with.
- Always consider all options and carefully compute the full solution when necessary! There will be options that seem correct but are unfortunately wrong.
- Make rational decisions! Expected score for uniform random choice: $1/5 * (+3 - 1 - 1 - 1 + 0) = 0$.
- Keep in mind the **type and variability of questions** from year to year! All questions are unique, and there are always unseen/new question types and variations.
- Our advice: Practice several of the exam with a strict 4 hour time limit, no aids condition and **without** the solutions.
- Do the practical exercises! They provide you with insights/intuitions into the methods and variations.

Rest of today... and beyond

Today

- No exercises session; TAs will be available from ~14:30 to answer questions about the course, previous exams, etc: **A081 and South lobby area** only!

Assistance beyond week 13: Piazza

- Please help each other (you can learn a lot from addressing questions from your fellow students)!
- Monitored/moderated **sporadically** by TAs/staff.
- Staff and TAs will not address/monitor new posts on Piazza after 16:00 on 27th May.

After the exam

- You can give feedback on the exam via DTU's systems...
- The answers (A, B,...) will be made available after the exam (~1-3 working days).
- We will may upload your Answer Sheet to DTU Learn (if enrolled on the course) with the detected answer. It will be announced **if/when** it happens.
- We cannot and will not inform **anyone** of their grades before they are published via the usual channels.

Thank you!

- And best of luck with the exam!